

Global Technology

AI Value Chain: How much does a GW of data center capacity actually cost, and what goes into it?



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Based on a series of conversations with industry experts as well as third-party data on physical infrastructure, we construct an estimate of rack-level economics for an GB200/ NVL72 AI data center. This note includes our analysis and key takeaways.

We estimate that a typical GB200 / NVL72 rack costs ~\$3.4M per rack. Coupled with ~\$2.5M in physical infrastructure costs per rack, this points to all-in AI data center capex of \$5.9M per rack or \$35B per GW. This estimate is notably lower than the \$50-60B number given by Nvidia on its Q2 2026 earnings call - we believe Nvidia is looking ahead to future product cycles.

Data center capex is notably dominated by the GPU, which we estimate at 39% of total costs, and by Nvidia gross profit dollars (29% of total costs). Given Nvidia's ~70% gross margins, this implies that Nvidia gross profit dollars account for ~30% of total AI data center spending. Even with lower ASIC margins, AI ASICs would remain the largest cost item by a significant margin: assuming COGS is the same for GPUs vs. ASIC but ASIC margins are 50% instead of 70%, this would take down the price of accelerated compute from ~\$2.3M to ~\$1.2M, a ~19% saving on total capex but still ~28% of ASIC data center capex.

Networking was the other big ticket item at ~13% of spend, although networking spend was also more dispersed across different types of equipment. Storage was relatively small at ~1.4% of spend.

How much foundry, HBM & WFE suppliers capture varies significantly between GPU & ASIC. With GB200, foundry gets 2.5-3% of data center capex (US\$1.1B/GW) & ~1% more if CPU (US\$0.3B/GW) too. HBM gets 3-3.5% (US\$1.1B/GW). WFE gets 3-4% (US\$1.2B/GW). With ASIC, they get much more, as the same data center capex can buy more chips.

The spending on mechanical & electrical equipment was less concentrated, but major items include diesel & gas generators and turbines (~6% of spend), uninterruptible power supplies (~4%) and transformers (~5%). Thermal management was a relatively small part of spend (~4%) and remained split between air cooling and liquid cooling, although we expect spend to continue to shift towards liquid cooling. We model in-rack power components content will increase to 7-8x in Rubin Ultra in 2027 with 800V HVDC design.

Given the shorter depreciation lifespan of IT Hardware such as servers and networking compared to mechanical & electrical equipment or land & buildings, and given that operating costs are relatively low, the true economic costs are likely even more heavily weighted towards servers and networking compared to what cash capex would imply. Even at an elevated cost of \$0.15/kWh, it costs ~\$1.3B in electricity to run a GW of data center capacity for a year. Personnel costs are also negligible, with 20 GW data centers reportedly operating with 8-10 people costing \$30-80k per year each.

BERNSTEIN TICKER TABLE

Ticker	Rating	24 Oct 2025		Price Target	TTM Rel. Perf.	Reported EPS				Adjusted P/E (x)		
		Cur	Closing Price			Cur	2024A	2025E	2026E	2024A	2025E	2026E
NVDA	O	USD	186.26	225.00	15.8%	USD	2.99	4.44	5.84	62.2	42.0	31.9
AVGO	O	USD	354.13	400.00	89.8%	USD	4.86	6.73	9.38	72.9	52.6	37.7
AMD	M	USD	252.92	200.00	47.9%	USD	3.31	3.92	5.27	76.4	64.5	48.0
INTC	M	USD	38.28	35.00	54.5%	USD	(0.13)	0.34	0.41	(289.5)	111.9	93.1
QCOM	O	USD	168.94	185.00	(16.4)%	USD	10.21	11.88	11.55	16.5	14.2	14.6
TXN	M	USD	169.13	160.00	(35.0)%	USD	5.20	5.44	5.72	32.5	31.1	29.6
ADI	M	USD	238.01	240.00	(13.3)%	USD	6.38	7.75	8.92	37.3	30.7	26.7
NXPI	M	USD	219.16	220.00	(25.2)%	USD	13.08	11.77	13.00	16.7	18.6	16.9
LRCX	O	USD	151.68	170.00	81.2%	USD	4.14	4.85	5.67	36.7	31.3	26.8
AMAT	O	USD	228.75	195.00	7.6%	USD	8.65	9.38	9.50	26.4	24.4	24.1
DELL	O	USD	158.64	180.00	14.7%	USD	8.10	9.55	11.11	19.6	16.6	14.3
2360.TT (Chroma ATE)	O	TWD	692.00	570.00	44.6%	TWD	12.41	19.40	21.77	55.8	35.7	31.8
2308.TT (Delta)	O	TWD	1,010.00	1,130.00	130.2%	TWD	13.53	23.25	28.73	74.7	43.4	35.2
2382.TT (Quanta)	U	TWD	293.50	240.00	(26.3)%	TWD	15.40	17.56	19.50	19.1	16.7	15.0
3037.TT (Unimicron)	O	TWD	156.00	170.00	(27.0)%	TWD	3.32	3.68	9.59	46.9	42.4	16.3
2330.TT (TSMC)	O	TWD	1,450.00	1,444.00	15.2%	TWD	45.25	60.94	70.54	8.8	7.3	6.0
TSM	O	USD	294.96	290.00	32.1%	USD	7.04	9.92	11.48	11.7	9.2	7.5
2454.TT (MediaTek)	O	TWD	1,295.00	1,600.00	(22.5)%	TWD	66.94	66.92	75.86	19.3	19.4	17.1
005930.KS (SEC)	O	KRW	98,900	95,000	55.9%	KRW	4,948.97	4,686.49	8,080.39	20.0	21.1	12.2
005935.KS (SEC-Pref)	O	KRW	77,900	80,750	42.5%	KRW	4,948.97	4,686.49	8,080.39	15.7	16.6	9.6
SMSN.LI	O	USD	1,726.00	1,734.00	46.3%	USD	90.66	83.78	147.45	19.0	20.6	11.7
000660.KS (SK hynix)	O	KRW	513,000	400,000	134.2%	KRW	28,300	50,488	64,734	18.1	10.2	7.9
MU	O	USD	219.02	170.00	88.1%	USD	1.26	8.14	14.64	173.4	26.9	15.0
ETN	O	USD	376.29	414.00	(7.9)%	USD	10.80	12.07	14.07	34.8	31.2	26.7
HUBB	O	USD	434.39	511.00	(20.4)%	USD	16.56	18.13	20.44	26.2	24.0	21.2
PWR	M	USD	440.93	410.00	25.4%	USD	8.97	10.70	13.58	49.1	41.2	32.5
LGN	O	USD	36.12	38.00	NA	USD	229.63	252.28	278.67	16.5	15.1	13.6
ASML	M	USD	1,033.10	935.00	28.4%	USD	20.83	27.00	28.18	42.7	32.9	31.5
8035.JP (Tokyo Electron)	O	JPY	30,270	31,100	6.2%	JPY	1,179.08	1,057.50	1,262.81	25.7	28.6	24.0
6525.JP (Kokusai)	O	JPY	4,877.00	5,580.00	53.7%	JPY	153.87	154.97	193.39	31.7	31.5	25.2
6146.JP (DISCO)	O	JPY	53,450	58,500	17.6%	JPY	1,119.74	1,153.84	1,389.59	47.7	46.3	38.5
6857.JP (Advantest)	M	JPY	17,080	9,430.00	94.2%	JPY	218.94	348.54	344.78	78.0	49.0	49.5
BESI.NA	O	EUR	147.05	169.00	19.6%	EUR	2.30	1.52	2.76	63.9	96.5	53.2
6920.JP (Lasertec)	U	JPY	21,680	10,900	(22.1)%	JPY	654.49	937.82	754.02	33.1	23.1	28.8
7735.JP (Screen)	M	JPY	13,375	12,000	16.1%	JPY	1,024.91	901.02	983.51	13.0	14.8	13.6
HPE	M	USD	23.50	24.00	4.2%	USD	1.98	1.88	2.69	11.9	12.5	8.7
SMCI	M	USD	48.29	46.00	(12.4)%	USD	2.04	2.10	2.85	23.7	23.0	17.0
STX	O	USD	234.12	275.00	108.3%	USD	8.10	11.36	14.47	28.9	20.6	16.2
WDC	M	USD	129.43	120.00	141.4%	USD	4.96	6.63	7.53	26.1	19.5	17.2
SNDK	O	USD	186.16	120.00	NA	USD	2.32	6.88	12.12	80.4	27.0	15.4
SPX			6,791.69									
ASIA			1,627.83									
EM			1,525.86									
JPL			2,119.17									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

NVDA, AVGO, AMD, INTC, QCOM, ADI, NXPI, LRCX, AMAT, MU, ETN, HUBB, PWR, HPE, SMCI, STX, WDC, SNDK estimate is Adjusted EPS; LGN estimate is EBITDA (M); TXN, 2360.TT, 2308.TT, 2382.TT, 3037.TT, 2454.TT, 005930.KS, 005935.KS, SMSN.LI, 000660.KS, ASML, 8035.JP, 6525.JP, 6146.JP, 6857.JP, BESI.NA, 6920.JP, 7735.JP valuation is Reported P/E (x); 2330.TT, TSM valuation is P/B (x); LGN valuation is EV/EBITDA (x); NVDA, LRCX, DELL, SMCI, STX, WDC, SNDK base year is 2025;

Source: Bloomberg, Bernstein estimates and analysis.

Ticker	Rating	24 Oct 2025		Price Target	TTM Rel. Perf.	Reported EPS				Adjusted P/E (x)		
		Cur	Closing Price			Cur	2024A	2025E	2026E	2024A	2025E	2026E
EDM			1,533.36									

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NVDA, AVGO, AMD, INTC, QCOM, ADI, NXPI, LRCX, AMAT, MU, ETN, HUBB, PWR, HPE, SMCi, STX, WDC, SNDK estimate is Adjusted EPS; LGN estimate is EBITDA (M); TXN, 2360.TT, 2308.TT, 2382.TT, 3037.TT, 2454.TT, 005930.KS, 005935.KS, SMSN.LI, 000660.KS, ASML, 8035.JP, 6525.JP, 6146.JP, 6857.JP, BESI.NA, 6920.JP, 7735.JP valuation is Reported P/E (x); 2330.TT, TSM valuation is P/B (x); LGN valuation is EV/EBITDA (x); NVDA, LRCX, DELL, SMCi, STX, WDC, SNDK base year is 2025;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

NVDA (OP, \$225): The datacenter opportunity is enormous, and still early, with material upside still possible.

AVGO (OP, \$400): A strong 2025 AI trajectory seems set to accelerate into 2026, bolstered by software, cash deployment, and superb margins & FCF

AMD (MP, \$200): AI expectations remain high, but a new deal with OpenAI has the prospect to drive further (possibly substantial) growth

INTC (MP, \$35): Intel's problems have broken through to the forefront.

QCOM (OP, PT \$185): AAPL headwinds are now here but are known, the product portfolio is stronger than ever, option value exists, and the shares are very inexpensive.

TXN (MP, \$160): TXN shares feel fully valued in the current environment.

ADI (MP, \$240): Pull-forward concerns exist amid strength, and shares are expensive.

NXPI (MP, \$220): The pace of recovery remains open for debate given tariff uncertainty (where auto maybe more adversely affected).

AMAT (OP, \$195): We maintain a positive view on secular WFE growth and see a number of drivers for AMAT including SAM growth, an increasing services narrative, and capital return.

LRCX (OP, \$170): The company is benefiting from key inflections (GAA, packaging, HBM, NAND upgrades) and CY26 commentary seems supportive

DELL (Outperform, \$180.00): The company continues to have large upside opportunities in AI servers and in storage, and is very inexpensive relative to its growth profile

SNDK (Outperform, \$120.00): After its spin-out from WDC, SNDK remains significantly undervalued and underappreciated as a leader in NAND, one of the core building blocks of the Intelligence Revolution.

STX (Outperform, \$275.00): Cyclical and structural tailwinds are aligning for STX, amid cyclical strength in HDDs and with STX's technology leadership in HAMR beginning to pay dividends.

WDC (Market-Perform, \$120.00): While WDC is inexpensive both in absolute terms and relative to STX, and will benefit from cyclical strength in HDDs, it will face both market share and margin pressure as it attempts to catch up in HAMR.

SMCI (Market-Perform, \$46.00): While Super Micro has been one of the stand-out growth stories in the story, its growth is decelerating sharply, guidance looks aggressive, and the business is more expensive than it looks given cash conversion challenges.

HPE (Market-Perform, \$24.00): HPE's transformational Juniper acquisition closed on July 2nd following a settlement with the DOJ, but we still struggle to underwrite the deal logic.

Samsung Electronics: We rate Samsung Electronics Outperform with price target of KRW95,000.

SK hynix : We rate SK hynix Outperform with price target of KRW400,000.

Micron: We rate Micron Outperform with price target of US\$170.00.

TSMC: We rate TSMC Outperform with price target of NT\$1,444.00.

MediaTek: We rate MediaTek Outperform with price target of NT\$1,600.00.

We rate **Tokyo Electron** (PT=¥31,100), **Kokusai** (PT=¥5,580.00), **DISCO** (PT=¥58,500), **Besi** (PT= EUR169.00) Outperform; **ASML** (PT= EUR935.00), **Advantest** (PT=¥9,430.00) , **Screen** (PT=¥12,000) Market-Perform, and **Lasertec** (PT=¥10,900) Underperform.

DETAILS

Based on a series of conversations with industry experts, supply chain checks, and third party data on out-of-rack costs of land and physical infrastructure, we construct an estimate breaking down rack-level economics for an GB200/NVL72 AI data center. This note includes our analysis and key takeaways.

We estimate that a typical GB200 / NVL72 rack costs ~\$3.4M per rack. Coupled with ~\$2.5M in physical infrastructure costs per rack, this points to all-in AI data center capex of \$5.9M per rack or \$35B per GW.

- **\$3.4M per rack for GB200 / NVL72.** Based on commentary from industry experts, we estimate the cost of a GB200 / NVL72 rack at \$2.9-\$3.5M, and arrive at ~\$3.4M for a “typical” rack (**Exhibit 1**).
- **\$35B per GW in total capex.** We estimate 132 kW in rack power consumption and 169 kW in total power consumption per rack (i.e. the rack is 78% of data center power). This points to 5.9k racks per GW, in turn implying \$20B in rack cost per GW. Industry contacts further pointed us to ~\$15B in out-of-rack physical infrastructure costs (translating to \$2.5M per rack), for an all-in data center capex of \$5.9M per rack or \$35B per GW (**Exhibit 2**).
- **What about Nvidia’s comments? Is that just “Jensen optimism”?** While the per GW opportunity for semis in general and Nvidia specifically is clearly huge, this estimate is notably lower than the \$50-60B/GW number given by Nvidia on its Q2 2026 earnings call, and more consistent with comments from Broadcom and AMD of ~\$15-20B in addressable opportunity per GW. Our industry contacts were explicit in calling out that they believed Nvidia’s number was too high, but we suspect Nvidia is looking ahead to future GPU cycles when per rack and per GW costs will be higher.

Data center capex is notably dominated by the GPU, which we estimate at 39% of total costs, and by Nvidia gross profit dollars (29% of total costs).

- **Nvidia gross profit dollars are the single largest line item in the AI data center BOM.** Given Nvidia’s ~70% gross margins, this implies that Nvidia gross profit dollars account for ~29% of total AI data center spending (**Exhibit 3**), almost as much as the entirety of the mechanical and electrical equipment (~33%). Despite capturing a relatively small share of GPU economics, the sheer size of GPU spend means that HBM, GPU silicon and other GPU costs remain relatively sizable line items in the overall BOM.
- **Even at lower ASIC margins, AI ASICs remain the largest cost item.** Assuming COGS is the same for GPUs vs. ASIC but ASIC margins are 50% instead of 70%, this would take down the price of accelerated compute from ~\$2.3M to ~\$1.2M, an ~19% saving on total capex and still ~24% of ASIC data center capex.
- **Are CPUs still relevant? More or less...** While at face value CPUs actually remain a fairly significant cost item (at 3% of total capex, its largely in line with components like switches which are more often argued to be AI winners). That said, given that CPUs are a small share of spend relative to GPUs and CPUs and GPUs are often bundled as superchips, CPUs are often seen as an add-on in the broader bundle.

Networking was the other big ticket item at ~13% of spend, although networking spend was also more dispersed across different types of equipment. Switches are the largest piece of networking equipment (~3% total spend), but other scale-up networking such as copper cabling (~2%) and backplane connectors like those supplied by Amphenol (~2%) as well as the scale-out networking infrastructure (e.g. network accelerators) were also significant.

- **Switches were ~3% of spending, with relatively high margins for both networking vendors and switch silicon designers.** While switches were not a massive cost item (likely 40-50% of scale-up networking, which in turn is 10-15% of rack costs), it was still fairly significant at ~3% of total spend. We also note that given networking vendors like Arista earn relatively high margins (60%+) on the switch hardware, and switch silicon designers are also earning relatively high margins on the silicon (which is the biggest component of COGS), switching is potentially more significant as a share of the profit pool than it is as a share of end-user spend.
- **DPU (3%), copper cabling (2%), backplane connectors (2%), NICS (2%) and transceivers (1%) were all also significant portions of spend.** Moreover, many of these categories have potential shifts in per-rack content, especially as co-packaged optics gains adoption.

Storage was a fairly small share of spend at ~1.4%. One of our industry contacts observed that a typical rack had ~2 PB of storage - at \$0.01/GB for hard disk drives, this would point even lower at ~\$20k per rack, or ~0.6% of rack cost. Notably, this recontextualizes why data center operators may prioritize other factors such as power efficiency over storage price per GB.

- **2PB of storage translates to ~0.6% of rack costs.** One of our industry contacts observed that a typical rack had ~2 PB of storage - at \$0.01/GB for hard disk drives, this translates to ~\$20k per rack, or ~0.6% of rack cost - which is actually lower than the industry experts feedback, potentially because some storage is on higher-cost NAND.
- **Have we been thinking about the QLC debate the wrong way? Does the HDD vs. NAND cost difference even matter?** This also recontextualizes the widely-debated [Meta white paper](#) in which Meta argued for more widespread use of QLC NAND flash in the data center instead of HDDs. Even with a significantly higher per GB cost (we estimate TLC SSD at \$0.066 and QLC is typically ~40% cheaper, so perhaps \$0.04), the incremental cost (perhaps \$60k per rack) is not very relevant compared to overall costs, and perhaps justifiable given that (as Meta argued), QLC SSD is more power efficient than either HDD and TLC SSD, which may be more important than the incremental cost savings in a power constrained environment.
- **What about the indirect spend?** That said, we observe that while direct AI data center spending on storage is relatively small, storage may also experience indirect benefits from AI. For instance, Dell and others have argued that enterprise AI adoption will require modernization of storage footprints, which in turn could drive a stronger refresh cycle in enterprise storage.
- **Further reading.** For Toni Sacconaghi's view on why storage is a relatively small part of spend, see: [Dell, HPE: Dimensioning the AI storage opportunity — Bonanza or wishful thinking?](#). For Mark Newman's primer on the HDD industry following his recent initiation on IT Hardware, see: [Global Hard Disk Drives: It's HAMR time! A primer on the HDD industry and why STX is poised to reap outsized rewards](#)

How much foundries, HBM & semicap and other upstream suppliers capture varies significantly between GPU & ASIC.

- **We estimate with GB200 the foundry opportunity at ~2.5-3% for GPU, and networking and the HBM memory opportunity at 3-3.5%, and WFE at 3-4%.** For the data centers deployed with GB200 NVL72, we estimate foundries get 2.5-3% of the data center capex through GPU, networking chips, etc. & ~1% more if CPU is also fabbed by foundries. Memory suppliers get 3-3.5% through HBM (regular DRAM is not included). WFE (wafer fab equipment) suppliers get 3-4%. This is equivalent to foundries getting US\$1.1B/GW, and another US\$0.3B/GW when CPU is also made by foundries. Memory companies will have US\$1.1B/GW through HBM & WFE suppliers will enjoy US\$1.2G/GW (**Exhibit 4**).
- **Lower price/margins and thus higher volumes implies that ASICs offer more favorable economics to foundries and other upstream suppliers.** However, if a data center is deployed with servers based on ASIC, low margins demanded by ASIC vendors implies more servers & hence more chips can be bought with the same amount of data center capex. Using the example above where ASIC vendors demand 50% gross margin vs. 70% enjoyed by NVIDIA, the data center capex can be lowered by ~16%, or equivalently 19% more racks/servers/chips can be purchased if the capex amount is unchanged. For the same reason, if ASIC is provided by Asian suppliers such as MediaTek or Alchip (3661 TT, not covered) who ask for gross margin below 50%, or GPU is from less demanding suppliers (AMD, & even Intel?), upstream suppliers like foundry, memory & WFE suppliers will get even more. The percentage numbers (as % of the data center capex) above hence should be scaled accordingly based on the savings from adopting these suppliers.
- **Power efficiency adds complexity.** The per-GW revenue estimates above however won't change if we simplistically assume GPU & ASIC differ in margins of suppliers but are equal in power efficiency. The reality is certainly more complex, and GPU & ASIC likely may optimize power efficiency for different use cases.
- **This is a snapshot in time... but how might it evolve?** We also suspect the estimate here is subject to technology evolution and may change over time, as shown in the trend from A100 to B300 in **Exhibit 5** for example. Our exercise here is based on a data center with GB200 NVL72, and should be adjusted to predict the future. We hence plan to address this, together with the changes from GPU to ASIC, in future research.
- **What about Lam's comment that it had an \$8B opportunity per \$100B in data center investment.** Some may find 3-4% above of data center capex going to WFE low, as Lam just guided US\$8B in WFE spending per US\$100B data center investment. We attribute the difference to that Lam's guidance likely assumes data center deployed with a mix of chips from

NVIDIA, AMD and ASIC suppliers. Lam's guidance may also embed some future changes that our analysis based on GB200 NVL72 doesn't capture. We also observe that Lam's guidance is about right if you assume that WFE spend is 20% of GPU spend (given GPUs are 40% of total spend). While the 20% number appears about right for many of the ASIC suppliers (**Exhibit 5**, again), it is high for Nvidia given Nvidia's higher capital efficiency (and conversely, ASICs are typically lower than 40% of total spend)

The spending on mechanical & electrical equipment was less concentrated, but major items include diesel & gas generators and turbines (~6% of spend), uninterruptible power supplies (~4%) and transformers (~5%). Thermal management was a relatively small part of spend (~4%) and remained split between air cooling and liquid cooling, although we expect spend to continue to shift towards liquid cooling (**Exhibit 2**, again).

- **Standby diesel and natural gas generators and turbines used for redundancy were the single largest piece of mechanical & electrical equipment at ~6% of spend.** These products include generators produced by Caterpillar (covered by Chad Dillard), MTU (owned by Rolls Royce, which is covered by Adrien Rabier), Kohler (not covered by Bernstein), and now Generac (not covered by Bernstein), who entered the market in April and started international deliveries in 2H25. While utility-scale power generation is out of scope of the analysis (since it does not sit within the data center and is instead reflected in power costs), they are increasingly in-focus for the debate - more on this later.
- **Power distribution was collectively ~10% of spend across a number of verticals and vendors.** Power distribution equipment makers include Eaton (covered by Chad Dillard), Schneider Electric (covered by Alasdair Leslie), Vertiv (uncovered), ABB (covered by Alasdair Leslie), nVent (uncovered), and Legrand (covered by Alasdair Leslie) to name a few. A greater repository of power distribution players can be found here: [Grid investment - who to play? A compendium of the 20 key electrical players](#). Increased electrical content is a key catalyst for players in this space. Eaton generates ~20% of revenues from data centers and distributed IT. In a traditional data center, Eaton can generate \$1.2-\$1.5M per MW in potential sales, representing 6-10% of total compute and infrastructure capex. In an AI data center, this opportunity can expand by 50% to \$1.2-\$2.9M in potential sales, up to 8% of total compute and infrastructure capex. For this reason, Eaton is currently investing \$1.5B to add capacity in 3-phase UPS, low- and medium-voltage assembly, switchboards and panelboards to satisfy incremental data center demand. It has also prioritized investing in modular construction via M&A (Fibrebond). Electrical component maker Hubbell has also entered the market through its acquisition of PCX.
- **Thermal management was relatively smaller at 4%.** Thermal management appeared to be a relatively small part of spend (~4%) and remained split between air and liquid cooling. That said, we expect spend to continue to shift towards liquid cooling. Moreover, players in the space like Vertiv (not covered by Bernstein) appeared more bullish on the opportunity than our analysis would suggest, with Vertiv citing a \$2.5-3M per MW opportunity for traditional compute applications that rises by ~20% to \$3-3.5M per MW for high-density compute applications.

Given the shorter depreciation lifespan of IT Hardware such as servers and networking compared to mechanical & electrical equipment or land & buildings, and given that operating costs are relatively low, the true economic costs are likely even more heavily weighted towards servers and networking compared to what cash capex would imply.

Even at an elevated cost of \$0.15/kWh, it costs ~\$1.3B in electricity to run a GW of data center capacity for a year. Personnel costs are also negligible, with 20 GW data centers reportedly operating with 8-10 people costing \$30-80k per year each.

By contrast, even on a 6 year depreciation cycle, which we see as generous¹, compute + networking costs ~\$3.4B in annual depreciation. Given the relatively high cost of capex relative to ongoing operating costs, and the fact that hardware has a shorter depreciation lifespan than physical infrastructure (**Exhibit 6**), the true TCO economics may actually be even more heavily weighted towards servers and networking compared to what cash capex would imply.

Looking forward, incremental economics are likely to accrue to companies that pose bottlenecks or which have potential for content increases. Beyond our present day snapshot of data center economics, we observe that companies which serve as a bottleneck are likely to capture an increasing share of economic value, as undersupply drives increased pricing and margins. Content increases also serve as an opportunity for companies to capture disproportionate economics. Lastly, we suspect that companies that do not serve as bottlenecks but which hold crucial technology or capabilities can manage supply in response to demand.

¹ It's true that GPUs don't explode at the end of three years. However, in a world where power is seen as the main bottleneck, it may not make sense to spend valuable power and data center space running 5-year old A100s that get 300 FP32 TFLOPS on 250 W when you could be running B200s that give you 2500 on 1000W.

- **Utility-scale power is increasingly a bottleneck to the space.** Industry participants increasingly cite the availability of powered land with commitments from grid utilities as a bottleneck to the continued build out. Utility-scale combined cycle gas turbine OEMs include GE Vernova (not covered), Mitsubishi Heavy (not covered), and Siemens Energy, who collectively own 70-80% of global production capacity and are expected to grow capacity from ~40 GWs in '25 to ~70 GWs by '27 to satisfy increased demand, particularly from data centers. For context, in its latest earnings call, Siemens Energy mentioned having 14 GWs of new orders YTD, 60% of which are coming solely from data centers.
- **Hyperscalars are looking for ways to circumvent this bottleneck.** Due to ongoing capacity constraints, hyperscalars are increasingly seeking alternatives via smaller industrial gas turbines from players like Caterpillar (covered by Chad Dillard), who owns Solar Turbines, the world's largest manufacturer of industrial gas turbines. Caterpillar's power generation capabilities also extend into standby diesel and natural gas generators used for redundancy. Other players include Cummins (covered by Chad Dillard), MTU (owned by Rolls Royce; covered by Adrien Rabier), Kohler (uncovered), and now Generac (uncovered) who entered the market in April and started international deliveries in 2H25. Alternative energy sources such as SMRs are also being widely explored as potential alternatives.
- **Could vendors with crucial capabilities and market cap take advantage? Possibly...** We observe that industry participants generally seem confident in 1-2 year demand visibility, likely because the pace of the AI build out is being dictated by power constraints and so AI labs are matching their build outs to the pace at which power becomes availability. While on the surface this would lend increased bargaining power to the power ecosystem, we wonder if vendors who own critical technologies, capabilities or market power can take advantage by then matching the pace of supply additions to the growth in demand. In doing so, vendors like TSMC or the memory vendors may be able to maintain a favorable demand-supply balance, helping to manage cyclicity in historically cyclical industries.
- **There is a significant opportunity for content increases in in-rack power. We model in-rack power content for Vera Rubin (VR200) will be 2-3x that of the GB200, and will increase to 7-8x in Rubin Ultra in 2027.** Near term, the power content increase in GB300 vs. GB200 is limited due to similar PSU and BBU spec, while a higher adoption rate of BBU could increase the power TAM by mid-teens to US\$46K (**Exhibit 7**). While the power design of VR200 is not finalized yet, we expect power content to at least double to US\$80-120K with higher power rate per rack and higher requirements for power stability (**Exhibit 8**), increasing the demand for BBU and super capacitors. The deployment of first-generation of **800V HVDC** (high voltage direct current) is likely to start in 2H26. Applying that to Bernstein volume estimates for Nvidia chips, we model power component TAM to increase by c. 70% next year and 30% in 2027 (**Exhibit 9**). More details can be found in [Delta Electronics: sizing the upside from power rack; PT raised to NT\\$1130](#).

For reference: Exhibit 10 includes a map of AI server supply chain. **Exhibit 11** includes companies in the AI server hardware supply chain. Detailed discussions of competition landscape in each sub-segment can be found in the following notes: [server ODM/OEMs](#), [power components](#), [liquid cooling](#), [copper connectors](#) and [optical transceivers](#)

EXHIBIT 1: **GB200 / NVL 72 Rack Capex Breakdown**

GB200 / NVL 72 BOM	\$K	\$B/GW	As % of fully loaded capex
CPU Silicon	85	0.5	1.4%
CPU Designer Gross Profit	85	0.5	1.4%
CPU	171	1.0	2.9%
HBM	192	1.1	3.2%
GPU Silicon ex. HBM	159	0.9	2.7%
GPU Designer Gross Profit	1,728	10.2	29.1%
Other GPU Costs (GPU Package Substrate, PCB, He	225	1.3	3.8%
GPU	2,304	13.7	38.8%
Computing	2,475	14.7	41.6%
Switch Silicon	13	0.1	0.2%
Switch Designer Gross Profit	53	0.3	0.9%
Networking Vendor Gross Profit	118	0.7	2.0%
Switches	184	1.1	3.1%
Copper Cabling	123	0.7	2.1%
Backplane Connectors	102	0.6	1.7%
Other scale-up networking	225	1.3	3.8%
Tranceivers	51	0.3	0.9%
NICS	102	0.6	1.7%
DPUs / Network Acceleration	188	1.1	3.2%
Scale-out networking	341	2.0	5.7%
Networking	751	4.5	12.6%
Power Delivery / Tray Chassis	34	0.2	0.6%
Power Distribution Nodes	17	0.1	0.3%
Intelligent Rack PDU		-	
Non-intelligent Rack PDU		-	
Open Compute Power Distribution		-	
Rack PDU Accessories		-	
Rack transfer switch		-	
Rack Power	51	0.3	0.9%
Liquid cooling	51	0.3	0.9%
Storage & Others	85	0.5	1.4%
Rack Total	3,413	20.2	57.4%

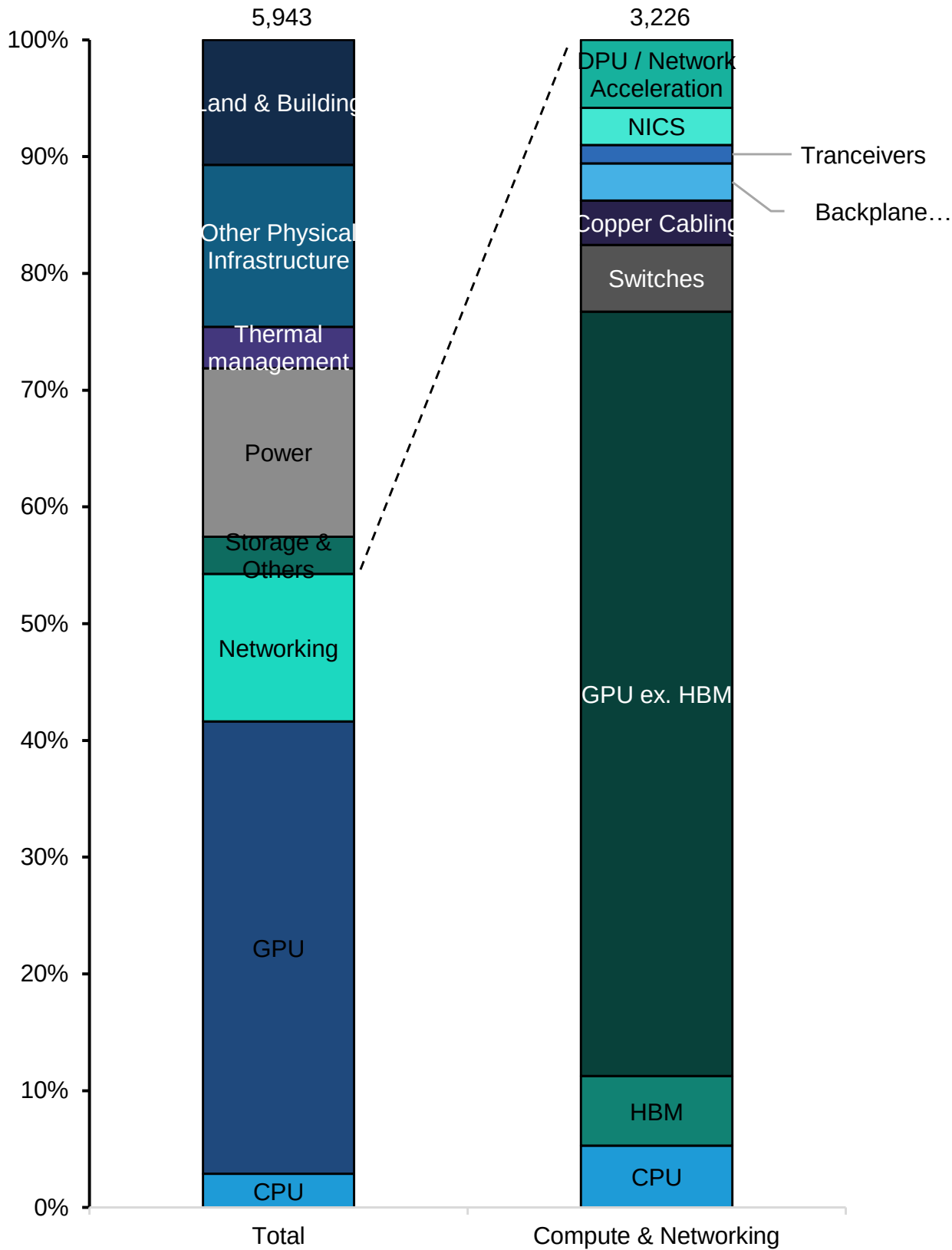
Assuming 70% gross margins for GPUs, 50% for CPUs, 64% for networking, 80% for switch silicon, and 70% for HBM
Source: Expert Conversations, Company Disclosures, Literature Search, Bernstein estimates and analysis

EXHIBIT 2: **Datacenter Infrastructure Capex Breakdown**

GB200 / NVL 72 BOM	\$K	\$B/GW	As % of fully loaded capex
Rack Total	3,413	20.2	57.4%
3P transformer-based PDU	19	0.1	0.3%
Busway	50	0.3	0.8%
Remote power panel	3	0.0	0.0%
Static transfer switch	7	0.0	0.1%
Cabling	93	0.5	1.6%
LV/MV Switchgear	110	0.7	1.9%
Transformer	306	1.8	5.1%
Power Distribution	587	3.5	9.9%
Battery backup unit (BBU)	14	0.1	0.2%
UPS Hardware	258	1.5	4.3%
Backup Power	272	1.6	4.6%
Air cooling	110	0.7	1.9%
Liquid cooling	44	0.3	0.7%
Supporting infrastructure	57	0.3	1.0%
Thermal management	211	1.3	3.6%
Diesel and gas generators & turbines	365	2.2	6.1%
Up Front DCIM software and sensor costs	55	0.3	0.9%
Physical Security	31	0.2	0.5%
Fire protection and suppression	18	0.1	0.3%
KVM Switch	15	0.1	0.2%
Ceiling and floor	11	0.1	0.2%
Lighting	10	0.1	0.2%
Other (e.g. pipe work, pumps, robots)	319	1.9	5.4%
Other Physical Infrastructure	824	4.9	13.9%
Mechanical and Electrical Total	1,894	11.2	31.9%
Land & Building	636	3.8	10.7%
Datacenter Capex Total	5,943	35.2	100.0%
Data Center Power to Support a Rack (kW)	169		
Rack Power (kW)	132		
Power Requirement Beyond Rack	78%		
Racks per GW	5,929		
Rack Cost per GW (\$B)	20		
Total Cost per GW (\$B)	35		

Source: Omdia, Expert Conversations, Company Disclosures, Literature Search, Bernstein estimates and analysis

EXHIBIT 3: GBL / NVL 72 Capex is dominated by the GPU and to a lesser extent networking

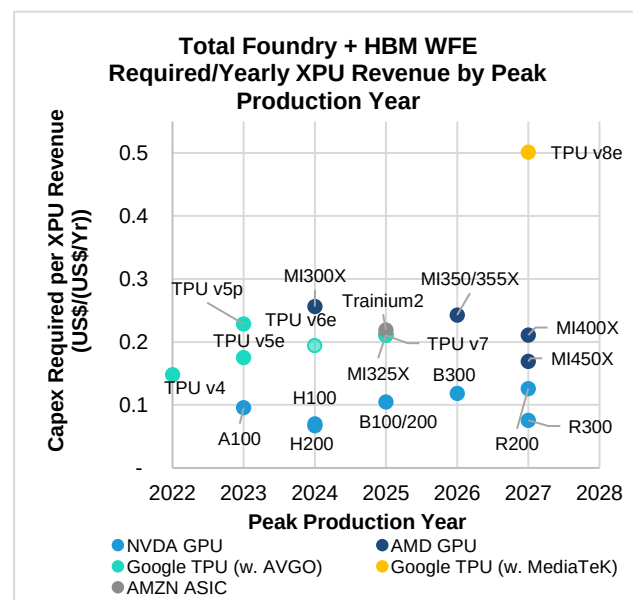


Source: Omdia, Expert Conversations, Company Disclosures, Literature Search, Bernstein estimates and analysis

EXHIBIT 4: How much foundries, HBM & other upstream suppliers capture varies significantly between GPU & ASIC.**Estimated Value Share in Data Center Capex**

Respective Value Share	Foundry		Memory (HBM only)	WFE
	XPU, Networking Chips etc.	CPU		
Using GB200 with 70% Gross Margin as an Example				
As % of Data Center Capex	2.5-3%	~1%	3-3.5%	3-4%
US\$/GW	1.1	0.3	1.1	1.2
Using AI ASIC with 50% Gross Margin as an Example				
As % of Data Center Capex	3-4%	~1.2%	3.5-4.5%	3.5-5%
US\$/GW	1.1	0.3	1.1	1.2

Source: Bernstein estimates and analysis

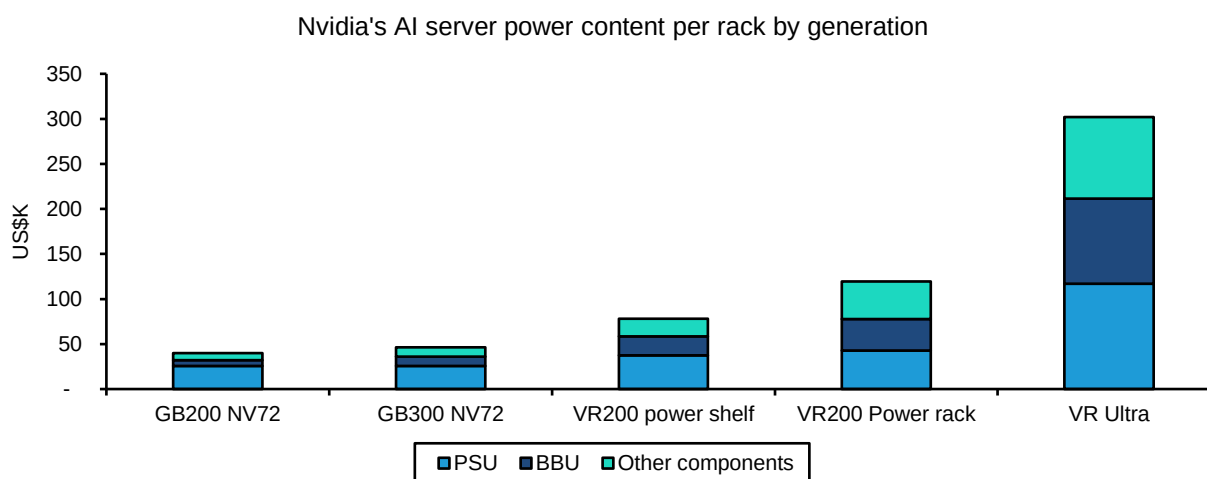
EXHIBIT 5: We suspect the WFE required to generate XPU value may change over time.

Source: Bernstein estimates and analysis

EXHIBIT 6: **Hyperscaler Disclosures on Depreciation Lifespans for Relevant Categories**

<u>Category</u>	<u>Company</u>	<u>Label</u>	<u>Disclosure</u>
IT Hardware	GOOGL	Servers & Network Equipment	Six Years
	AMZN	Servers and Networking Equipment	Five Years
	META	Servers and network assets	Four to Five years (Effective J
	MSFT	Computer Equipment	Two to Six Years
	ORCL	Computer, network, machinery and equipm	1-6 years
	NBIS	Server and network equipment	4.0 years
	CRWV	Technology Equipment	6 years
Mechanical & Electrical	GOOGL		
	AMZN		
	META	Equipment and other	One to 25 years
	MSFT	Furniture and equipment	One to 10 years
	ORCL		
	NBIS	Infrastructure systems and equipment	3.0-10.0 years
	CRWV	Data center equipment	8-12 years
Land & Building	GOOGL	Data Center and Office Buildings	7 to 40 years
	AMZN		
	META	Buildings	25 to 30 years
	MSFT	Buildings and Improvements	Five to 15 years
	ORCL	Buildings and Improvements	1-40 years
	NBIS	Buildings	20.0 years
	CRWV		

Source: Company Filings, Bernstein analysis

EXHIBIT 7: **We expect VR Ultra's power content to grow by 7.6x vs. GB200**

BBU's content factored in adoption rate
Source: Bernstein estimates and analysis

EXHIBIT 8: Power component per rack in Nvidia's AI server

Power content in Nvidia AI server racks					
	GB200 NV72	GB300 NV72	VR200 power shelf	VR200 Power rack	VR Ultra
GPU TDP (kw)	1.2	1.4	2	2	3.6
# GPU	72	72	72	72	144
GPU TDP per rack (kw)	86	101	144	144	518
Other chips & components TDP per rack (CPUs, Networkin	44	44	56	76	82
Server power consumption excl. redundancy (kw)	130	145	200	220	600
Design power capacity per rack	198	198	288	330	900
	1.5	1.4	1.4	1.5	1.5
PSU					
Power spec	6*33kw	6*33kw or 4*72kw	4*72kw?	10*33kw?	5*180kw?
PSU ASP per kw (US\$)	130	130	130	130	130
PSU kw/unit	5.5	5.5	12.0	5.5	30.0
PSU ASP per unit (US\$)	715	715	1,560	715	3,900
Total PSU content per rack (US\$K)	26	26	37	43	117
BBU					
BBU ASP per kw (US\$)	105	105	105	105	105
BBU spec per unit (kw)	5.5	5.5	12.0	5.5	30.0
BBU ASP per unit (US\$)	578	578	1,260	578	3,150
Total BBU content per rack (US\$K)	21	21	30	35	95
Adoption rate	30%	50%	70%	100%	100%
BBU content * adoption rate (US\$K)	6	10	21	35	95
Others					
Other power/mechanical component cost (US\$K)	8	10	20	42	91
% other power component	20%	22%	25%	35%	30%
Total power component per rack (US\$K)	40	46	78	119	302
vs. GB200		1.16x	2.0x	3.0x	7.6x

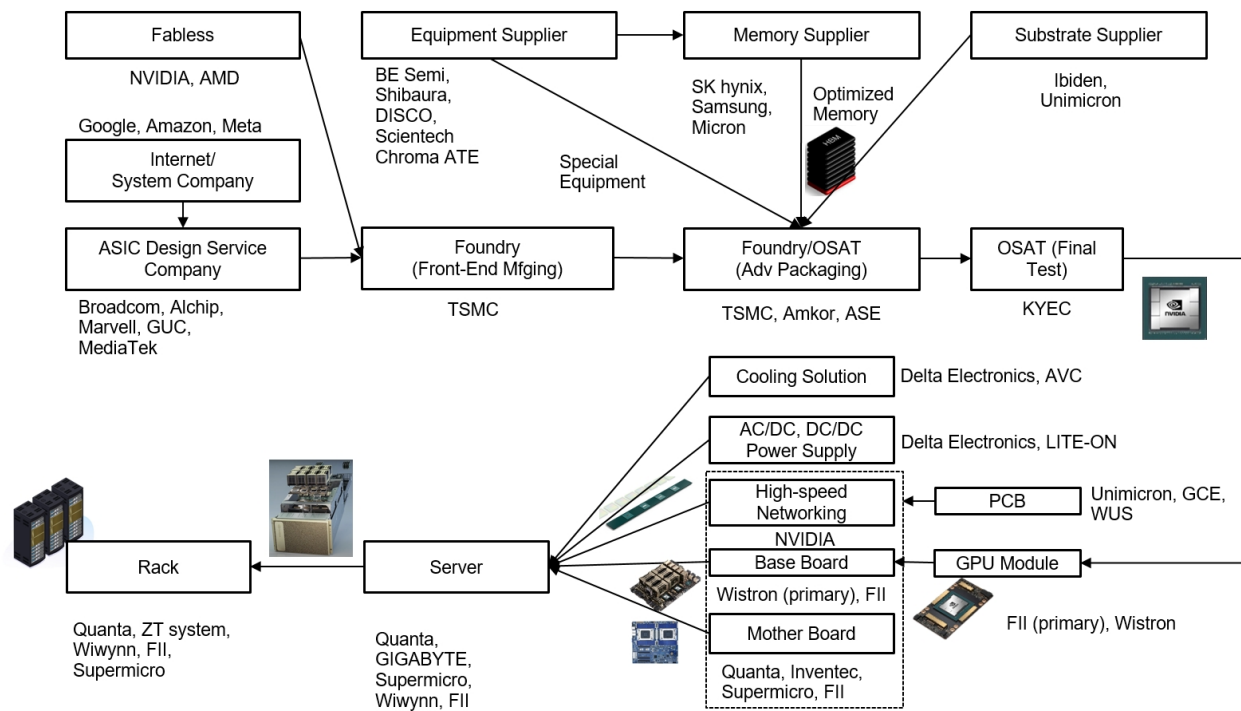
Source: Delta, Bernstein estimates and analysis

EXHIBIT 9: Nvidia's AI server power component TAM

Power component TAM for Nvidia racks	2025E	2026E	2027E
Nvidia chip shipment (K)	6,629	8,087	5,829
B200	4,349	2,507	80
B300	2,280	3,455	968
R200		2,124	3,181
R300			1,600
# chips per rack			
B200/B300/R200	72	72	72
R300			144
Nvidia GPU rack shipment (K)	91	110	69
GB200 rack	60	34	1
GB300 rack	31	47	13
VR200 rack		29	44
VR Ultra rack			11
Total power component content per rack (US\$M)			
GB200/B200A rack	40		
GB300 rack	46		
VR200 rack	99		
VR Ultra rack	302		
Blended power component content in each rack shipped yoy%	42	58	120
		38%	107%
Power component TAM (US\$M)			
GB200 rack	2,398	1,359	40
GB300 rack	1,436	2,177	602
VR200 rack		2,863	4,344
VR Ultra rack			3,324
Total power component TAM (US\$M)	3,834	6,399	8,310
yoy%		67%	30%

Source: Bernstein estimates and analysis

EXHIBIT 10: AI Server Supply Chain



Note: Bernstein covers Quanta, Delta Electronics, Unimicron, Chroma ATE, TSMC, Samsung, SK hynix, Micron, NVIDIA, AMD, Broadcom, DISCO, Ibiden, BE Semi, but not the rest.

Source: Nvidia, SK hynix, Bernstein analysis

EXHIBIT 11: **Summary of AI supply chain companies**

Position in Supply Chain	Ticker	Company	Mkt Cap (US\$M)	Forward P/E	Revenue (US\$M)		GPM %		EPS
					2024	24-26 CAGR	2024	2026E	24-26 CAGR
OEM	DELL US Equity	Dell 	104,265	14.2x	95,567	11%	22.8%	20.2%	18%
	SMCI US Equity	SuperMicro 	28,478	16.8x	21,972	36%	11.2%	10.7%	29%
	HPE US Equity	HPE 	30,598	9.7x	30,127	16%	32.8%	33.1%	10%
ODM	2317 TT Equity	Hon Hai 	110,160	14.9x	213,554	22%	6.3%	5.9%	25%
	601138 CH Equity	FII 	178,397	29.9x	84,606	45%	7.3%	6.5%	43%
	2382 TT Equity	Quanta 	36,377	14.2x	43,920	45%	7.9%	6.0%	20%
	6669 TT Equity	Wiwynn 	24,575	15.6x	11,224	76%	10.4%	7.5%	49%
	3231 TT Equity	Wistron 	14,680	13.0x	32,665	59%	8.0%	5.7%	41%
Copper connectivity	APH US Equity	Amphenol 	165,203	35.6x	15,223	31%	33.8%	37.4%	44%
	002475 CH Equity	Luxshare 	61,481	23.0x	37,334	21%	10.4%	11.6%	25%
Optical transceiver	300308 CH Equity	Innolight 	68,773	37.7x	3,314	51%	33.8%	38.6%	70%
	300502 CH EQUITY	Eoptolink 	48,528	29.6x	1,201	100%	44.7%	46.8%	116%
	300394 CH Equity	TFC Optical 	18,248	45.5x	452	56%	57.2%	52.8%	54%
	COHR US Equity	Coherent 	19,071	24.5x	5,810	10%	37.9%	39.9%	28%
Power / thermal component	2308 TT Equity	Delta 	86,190	35.1x	13,111	25%	32.4%	35.6%	53%
	VRT US Equity	Vertiv 	70,044	37.5x	8,012	23%	36.6%	37.4%	33%
	FLEX US Equity	Flex 	23,980	19.9x	25,813	4%	8.4%	9.3%	12%
	3017 TT Equity	AVC 	14,838	20.8x	2,234	54%	23.5%	26.7%	71%
	2301 TT EQUITY	Lite-On 	12,624	22.1x	4,269	20%	22%	23%	27%

Source: Bernstein covers Dell, SuperMicro, HPE, Quanta, Luxshare, Delta, but not the rest

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